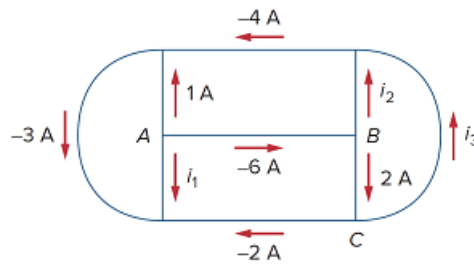


Problem

Find i_1 , i_2 , and i_3 in Fig. 2.73.

Figure 2.73:



Step-by-step solution

Step 1 of 2

Refer to the circuit diagram in Figure 2.73 in the textbook.

Apply Kirchhoff's current law at node A.

$$\begin{aligned} -1 - i_1 - (-6) &= 0 \\ -1 \text{ A} - i_1 + 6 \text{ A} &= 0 \\ -i_1 + 5 \text{ A} &= 0 \\ i_1 &= 5 \text{ A} \end{aligned}$$

Step 2 of 2

Apply Kirchhoff's current law at node B.

$$\begin{aligned} -6 \text{ A} - i_2 - 2 \text{ A} &= 0 \\ -i_2 - 8 \text{ A} &= 0 \\ i_2 &= -8 \text{ A} \end{aligned}$$

Apply Kirchhoff's current law at node C.

$$\begin{aligned} 2 \text{ A} - (-2 \text{ A}) - i_3 &= 0 \\ 2 \text{ A} + 2 \text{ A} - i_3 &= 0 \\ i_3 &= 4 \text{ A} \end{aligned}$$

Therefore, the required currents are,

$$\begin{aligned} i_1 &= 5 \text{ A} \\ i_2 &= -8 \text{ A} \\ i_3 &= 4 \text{ A} \end{aligned}$$

[Comments \(2\)](#)

☐ **Anonymous**

wow

☐ **Anonymous**

how do I know which current is negative or positiv?